3-4 Information Technology for Safety [1]

Point!

1 Encryption

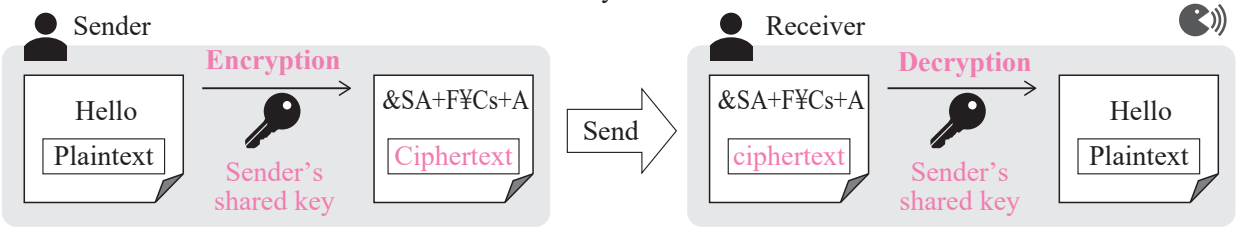
(1) ([1]**Encryption**) : A method used when sending information to prevent it from being intercepted by anyone other than the intended recipient. The encrypted text is called ([2]**ciphertext**), and the original, unencrypted text is called ([3]**plaintext**).

(2) ([4]**Decryption**) : The process of converting ciphertext back into its original plaintext form.

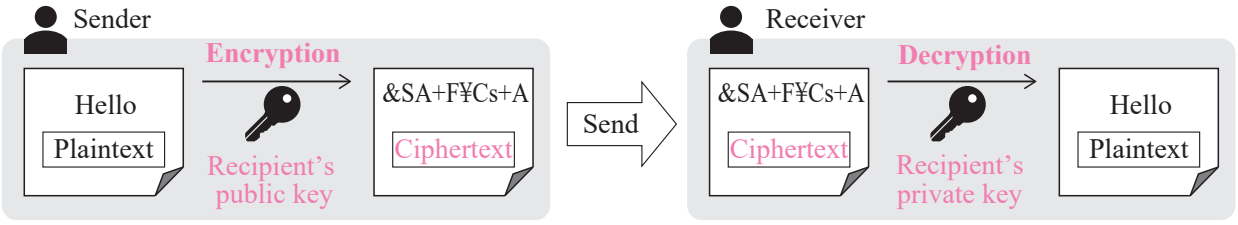
(3) ([5]**Key**) : The specific procedure or data used for encryption and decryption.

2 Types of encryption

(1) ([6]**Symmetric key encryption**) : An encryption method where the same ([7]**shared key**) is used for both encryption and decryption.
The message is encrypted using the ([8]**sender's shared key**) and decrypted using the ([9]**sender's shared key**) that was sent in advance by the sender.



(2) ([10]**Public key encryption**) : An encryption method that uses a publicly shared encryption key ([11]**public key**) and a private encryption key ([12]**private key**). The message is encrypted using the ([13]**recipient's public key**), which was sent in advance, and decrypted using the ([14]**recipient's private key**), which only the recipient possesses.



(3) Characteristics of Symmetric Key Encryption and Public Key Encryption

	Symmetric Key Encryption	Public Key Encryption
Merit	Compared to public-key encryption, encryption and decryption processing speed is ([15]**faster**).	Since data can be decrypted by anyone with the key, a different shared key is needed for each sender.
Demerit	Because the public key can be shared freely, key management is easier.	Compared to symmetric-key encryption, encryption and decryption processing speed is ([16]**slower**).

(4) ([17]**Session key method**) : An encryption method that combines symmetric key encryption and public key encryption.

Warm Up

Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms.

> When sending information, the technology used to prevent it from being leaked or tampered with by anyone other than the intended recipient is called ([1]). The original data before encryption is called ([2]), and the act of converting the ciphertext back into plaintext is called ([3]). Also, during both ([1]) and ([3]), something called a ([4]) is used.

(2) For each of the following items **A** to **D**, write "S" if the statement refers to symmetric key encryption, or "P" if it refers to public key encryption.

A The encryption key is made public, and encryption is done using the public key while decryption is done with the private key.

B Encryption uses separate keys held by the recipient — one for encryption and one for decryption.

C Compared to the other method, the encryption and decryption processing speed is faster.

D Compared to the other method, exchanging the key is more difficult.

(3) What is the name of the hybrid encryption method that combines symmetric key encryption and public key encryption?

Explanation

(1) [1] encryption [2] plaintext [3] decryption [4] key

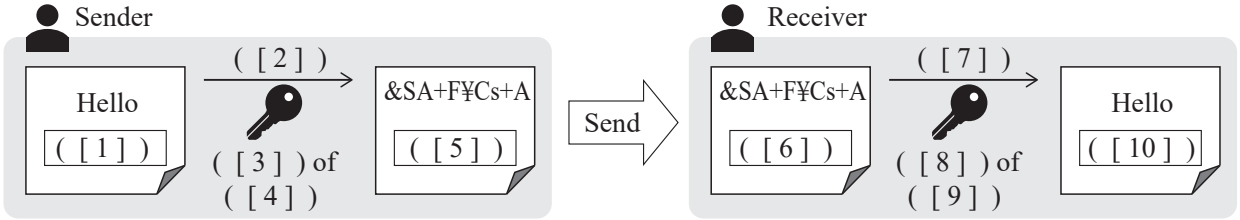
(2) **A** and **B**: The method in which encryption is performed with the recipient's public key and decryption with the recipient's private key is public key encryption.
C and **D**: Symmetric key encryption is faster than public key encryption, but since it uses the same key for both encryption and decryption, there is the issue of how to securely share the key with the recipient.
Therefore, A: **P**, B: **P**, C: **S**, D: **S**

(3) Session key method

Try

1 Answer the following question.

(1) The diagram illustrates the process flow of public key encryption. For blanks [1] to [10], choose the appropriate terms from options **A** to **J** below. Note that the same option may be used more than once.



A Plaintext **B** Ciphertext **C** Decryption **D** Encryption **E** Encoding
F Shared key **G** Public key **H** Private key **I** Sender **J** Recipient

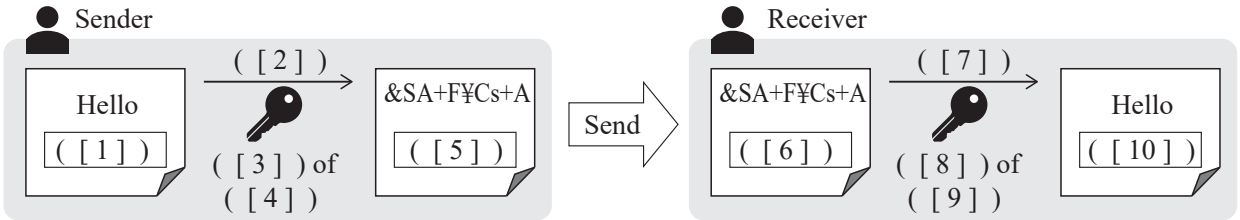
(2) For each of the following items **A** to **D**, write "S" if the statement refers to symmetric key encryption, or "P" if it refers to public key encryption.

A A separate key must be prepared for each sender.
B The same key is used by the sender for both encryption and decryption.
C Encryption and decryption are slower compared to the other method.
D Exchanging the key is easier compared to the other method.

Exercise

1 Answer the following question.

(1) The diagram shows the process flow of symmetric key encryption. For blanks [1] to [10], choose the appropriate terms from the options **A** to **J** below. The same option may be used more than once.



A Plaintext **B** Ciphertext **C** Decryption **D** Encryption **E** Encoding
F Shared key **G** Public key **H** Private key **I** Sender **J** Recipient

(2) Among the options **A** to **D** below, choose the one that best describes a characteristic of symmetric key encryption when compared to public key encryption. Answer using the corresponding letter.

A Uses different keys for encryption and decryption.
B Allows fast encryption and decryption.
C Enables safer distribution of keys.
D Makes key management easier even when communicating with many different parties.